

Study Questions: Action

1. When you work 2 sc into a single stitch (an increase), you add surface area without adding boundary length. Explain how this creates **negative Gaussian curvature** and why the fabric ruffles.
2. A flat circle in single crochet requires exactly 6 increases per round. What happens if you consistently work 8 increases per round instead? Describe the resulting geometry in terms of Gaussian curvature.
3. Compare a single crochet decrease (sc2tog) to a neural network reducing a weight during gradient descent. In what sense are both operations "removing" something to change the shape of a landscape?
4. Describe the topological difference between working in rows (back and forth) and working in joined rounds. What happens to the fabric's boundary and sidedness when you join the last stitch of a round to the first?
5. When you skip a stitch and chain over the gap in a lace pattern, you create a hole. How does this change the **genus** of the crocheted surface? Why does this matter topologically?
6. Explain the difference between working into the front loop only (FLO), back loop only (BLO), and both loops. How does each option change the structural connectivity between the current row and the row below?
7. A crocheter making an amigurumi sphere uses increases to expand, straight rounds to maintain, and decreases to close. Map each phase to its Gaussian curvature (positive, negative, or zero) and explain why.
8. In neural network training, **backpropagation** traces error backward through the layers to find which weights need adjustment. Describe how a crocheter traces a shaping error (a ruffle or pucker appearing three rounds later) back to the round where the increase or decrease count was wrong.

9. Frogging (ripping back) is the crocheter's way of rolling back gradient steps. Under what circumstances is frogging the better choice than compensating for the error in the next round? Frame your answer in terms of expected free energy.
10. A crocheted Mobius strip is made by joining a strip with a half-twist. Why can this twist not be removed by stretching, blocking, or reshaping? What makes it a genuinely topological feature rather than a geometric one?
11. In active inference, an agent selects actions to minimize the difference between its predictions and observations. Describe a specific moment in crochet where the crocheter perceives a mismatch between the predicted shape and the actual shape, and then selects an action to correct it.
12. The **action space** of a crocheter includes stitch type, placement, tension, and the decision to continue or frog. How does this multidimensional action space compare to the weight space of a neural network? In what ways is the crocheter's space richer or more constrained?
13. Daina Taimina's crocheted hyperbolic planes are made by increasing in every stitch (or every other stitch). Why does a constant increase ratio produce a surface of constant negative curvature? What would happen if the increase ratio changed from round to round?
14. A garment designer shapes a waist by decreasing over several rounds, then increasing again at the hips. Describe this shaping sequence as a curvature profile — positive curvature, then zero, then negative — and explain how it matches the three-dimensional topology of the human torso.
15. Compare skip connections in a neural network (where information jumps over layers) to a crocheter working a stitch into a row several rows below (a surface crochet technique). How do both operations change the connectivity topology of their respective systems?
16. When a crocheter joins a flat circle into a tube by working straight rounds (no increases), the topology changes from a disc to a cylinder. What additional topological operation must they perform to create a closed sphere?

17. Explain how the crocheter's **generative model** (the pattern or mental image of the finished object) guides action selection at each stitch. What happens when the generative model is vague or incomplete — for example, when improvising without a pattern?
18. In a neural network, the **loss landscape** can have local minima — configurations that seem good locally but are not globally optimal. Describe a crochet situation that parallels a local minimum: a shaping decision that seems right at the moment but leads to problems later.
19. Mathematical crochet projects (hyperbolic planes, Lorenz manifolds, coral reef sculptures) follow computed algorithms to control curvature. How does the precision required by these mathematical patterns differ from the looser curvature control in everyday amigurumi? What does this tell us about different levels of precision in generative models?
20. Reflect on a shaping decision you have made in your own crochet practice — an increase, a decrease, a join, or a frogging episode. Describe it as a topological operation and as an active inference action. What was the prediction, what was the observation, and what action did you select?